

Game Theory-Assignment2

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Exercise 3

Consider the zero-sum matrix game

$$A = \begin{pmatrix} -1 & -3 & -3 & 1 \\ 0 & 1 & -2 & -1 \\ 2 & -2 & 0 & -1 \end{pmatrix},$$

where player 1 (the minimizer) selects rows and player 2 (the maximizer) selects columns. Player 1 moves first.

(1)

Player 1 evaluates the worst-case payoff of each row:

$$\begin{aligned} \max_j A_{1j} &= 1, \\ \max_j A_{2j} &= 1, \\ \max_j A_{3j} &= 2. \end{aligned}$$

Thus, player 1 minimizes the worst-case outcome by choosing either row 1 or row 2, each guaranteeing a payoff of at most 1.

A rational policy for player 1 is therefore:

Choose row 1 or row 2.

Given the observed action of player 1, player 2 responds optimally:

- If player 1 chooses row 1, player 2 chooses column 4.
- If player 1 chooses row 2, player 2 chooses column 2.
- If player 1 chooses row 3, player 2 chooses column 1.

(2)

Against the policy of player 2, player 1 obtains payoff 1 when choosing either row 1 or row 2, and payoff 2 when choosing row 3. Hence, row 3 is strictly worse, and rows 1 and 2 are best responses.

Against the policy of player 1, player 2 always selects a column that maximizes the payoff for the realized row. Therefore, neither player can improve their payoff by unilaterally deviating from their policy, and the pair of policies is regret-free.

Exercise 4

Matrix A

$$A = \begin{pmatrix} 2 & -1 & 1 & -1 \\ -2 & -3 & 1 & -2 \\ -1 & -2 & -3 & -4 \\ 1 & -1 & 0 & -1 \end{pmatrix}$$

The row security levels are

$$\min_j A_{ij} = (-1, -3, -4, -1),$$

so the security level for player 1 is

$$v_1 = \max_i \min_j A_{ij} = -1.$$

The security policies for player 1 are rows 1 and 4.

The column security levels are

$$\max_i A_{ij} = (2, -1, 1, -1),$$

so the security level for player 2 is

$$v_2 = \min_j \max_i A_{ij} = -1.$$

The security policies for player 2 are columns 2 and 4.

Pure saddle points are given by the intersections of security policies. The pure saddle-point equilibria are

$$(1, 2), (1, 4), (4, 2), (4, 4).$$

Matrix B

$$B = \begin{pmatrix} 1 & 1 & -1 \\ -2 & 1 & -2 \\ 1 & 1 & -1 \end{pmatrix}$$

The row security levels are

$$\min_j B_{ij} = (-1, -2, -1),$$

so the security level for player 1 is

$$v_1 = -1,$$

with security policies rows 1 and 3.

The column security levels are

$$\max_i B_{ij} = (1, 1, -1),$$

so the security level for player 2 is

$$v_2 = -1,$$

with security policy column 3.

The pure saddle-point equilibria are

$$(1, 3), (3, 3).$$

Exercise 5

(1) No pure saddle-point equilibria

$$\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

No entry is simultaneously the minimum of its row and the maximum of its column.

(2) A single pure saddle-point equilibrium

$$\begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix}$$

The unique saddle point is at entry $(1, 1)$.

(3) Exactly two pure saddle-point equilibria

$$\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$

Both entries in the first row are saddle points.

(4) Exactly three pure saddle-point equilibria

Such a configuration is not possible in a 2×2 zero-sum game, since saddle points must form a Cartesian product of security strategies.

(5) Exactly four pure saddle-point equilibria

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

All four entries are saddle points.